

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0909

Sheepskin 100 kV
February 24, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0909. The Interconnection Customer proposes a 212.7 MW solar photovoltaic facility interconnecting at the Sheepskin 100 kV substation in the Indiana area. The request was received on December 28, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 237 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$63,627,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Caguanas 8103 - Columbia Hills 100 kV
Contingency: Lonsdale - Sheepskin 100 kV
Loading: 219.5% (Rating: 89 MVA)
DFAX: 100.0%
Pre-project loading: 19.4%
2. Facility: Caguanas 8103 - Sheepskin 100 kV
Contingency: Lonsdale - Sheepskin Substation 100 kV
Loading: 187.4% (Rating: 109 MVA)
DFAX: 100.0%
Pre-project loading: 8.2%
3. Facility: Lonsdale - Sheepskin 100 kV line
Contingency: Caguanas 8103 - Sheepskin 100 kV
Loading: 171.9% (Rating: 119 MVA)
DFAX: 100.0%
Pre-project loading: 7.5%
4. Facility: Lonsdale - Hubbard 100 kV
Contingency: Richland - Dekalb 500 kV
Loading: 119.8% (Rating: 161 MVA)
DFAX: 19.7%
Pre-project loading: 93.8%
5. Facility: China - Ponnequin 100 kV
Contingency: Wabash - Montpelier Electric Generating Sta 500 kV
Loading: 118.2% (Rating: 119 MVA)
DFAX: 6.0%
Pre-project loading: 107.5%

6. Facility: Hillcrest - Muldoon Mill 100 kV line
Contingency: Wayne Trace - Darlington Road 100 kV
Loading: 118.2% (Rating: 319 MVA)
DFAX: 5.7%
Pre-project loading: 114.3%
7. Facility: Bedford Rural - China 100 kV
Contingency: Wabash Substation - Montpelier Electric Generating Station 500 kV line
Loading: 116.6% (Rating: 108 MVA)
DFAX: 6.0%
Pre-project loading: 104.8%
8. Facility: Wabash 500/230 kV TX
Contingency: Richland - Dekalb 500 kV
Loading: 115.4% (Rating: 1,022 MVA)
DFAX: 32.9%
Pre-project loading: 108.6%
9. Facility: Adirondack Substation 500/230kV Auto
Contingency: East Methuen Substation 500/230 kV Auto
Loading: 114.1% (Rating: 1,292 MVA)
DFAX: 7.8%
Pre-project loading: 112.8%
10. Facility: Waynedale - Montpelier Electric Generting Sta 230 kV
Contingency: Richland - Dekalb Substation 500 kV
Loading: 112.6% (Rating: 1,036 MVA)
DFAX: 15.5%
Pre-project loading: 109.4%
11. Facility: Troy Energy Llc - Trego 500 kV
Contingency: Swan Vly - Trego 500 kV line
Loading: 112.1% (Rating: 4,063 MVA)
DFAX: 19.2%
Pre-project loading: 111.1%
12. Facility: James De Young - Adirondack 230 kV Line
Contingency: St. Lawrece - James De Young 230 kV
Loading: 110.9% (Rating: 1,172 MVA)
DFAX: 8.5%
Pre-project loading: 109.4%
13. Facility: Pickford - Herald-Twin Cities 230 kV L
Contingency: Trego - Beaver Substation 500 kV
Loading: 110.7% (Rating: 1,661 MVA)
DFAX: 8.2%
Pre-project loading: 109.6%
14. Facility: Gray'S Flat Substation - East Methuen 230 kV
Contingency: Madeline - E Methuen 230 kV
Loading: 110.5% (Rating: 1,373 MVA)
DFAX: 10.7%
Pre-project loading: 108.9%
15. Facility: Mason St - New Departure 230 kV
Contingency: Trego - Beaver 500 kV
Loading: 110.5% (Rating: 1,300 MVA)
DFAX: 6.4%
Pre-project loading: 109.4%
16. Facility: Adams - Onancock Diesel Generation Facility 230 kV
Contingency: Beavr - Trego 500 kV

- Loading: 110.4% (Rating: 2,527 MVA)
DFAX: 13.5%
Pre-project loading: 109.3%
17. Facility: Pickford - Adams 230 kV
Contingency: Trego - Beaver 500 kV
Loading: 110.3% (Rating: 1,282 MVA)
DFAX: 7.1%
Pre-project loading: 109.2%
18. Facility: Trego - Beaver 500 kV line
Contingency: Adams - Onancock Diesel Generation Facility 230 kV
Loading: 109.6% (Rating: 5,059 MVA)
DFAX: 28.1%
Pre-project loading: 108.4%
19. Facility: Beaver 230/1kV Transformer
Contingency: Beaver - Eufaula 500 kV line
Loading: 108.6% (Rating: 2,684 MVA)
DFAX: 13.1%
Pre-project loading: 107.5%
20. Facility: Beaver 500 kV/1 kV XFMR
Contingency: Beaver - Eufaula 500 kV
Loading: 108.6% (Rating: 2,684 MVA)
DFAX: 13.1%
Pre-project loading: 107.5%
21. Facility: Gavin Power Llc 500-345 kV Transformer
Contingency: Olympus 500-230 kV Auto
Loading: 108.5% (Rating: 1,310 MVA)
DFAX: 5.3%
Pre-project loading: 107.6%
22. Facility: Myrtle Pt - Georg Neal South 230 kV Line
Contingency: Trego - Galion Generating Sta 500 kV
Loading: 108.1% (Rating: 974 MVA)
DFAX: 6.6%
Pre-project loading: 106.7%
23. Facility: Gruver - Galeville 230 kV
Contingency: Imperial - Eufaula 500 kV Line
Loading: 108.1% (Rating: 753 MVA)
DFAX: 5.8%
Pre-project loading: 106.4%
24. Facility: Canal Road - Weedpatch 230 kV L
Contingency: Beaver - Trego 500 kV Line
Loading: 108.0% (Rating: 1,187 MVA)
DFAX: 7.5%
Pre-project loading: 106.7%
25. Facility: Hubbard - Wrangell 100 kV Line
Contingency: Richland - Dekalb 500 kV
Loading: 107.9% (Rating: 159 MVA)
DFAX: 19.7%
Pre-project loading: 81.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Caguanas 8103 - Columbia Hills 100 kV	Rebuild line with higher-capacity conductor	\$4,739,000	100% (\$4,739,000)
Caguanas 8103 - Sheepskin 100 kV	Rebuild line with higher-capacity conductor	\$5,004,000	100% (\$5,004,000)
Lonsdale - Sheepskin 100 kV	Rebuild line with higher-capacity conductor	\$29,682,000	100% (\$29,682,000)
Lonsdale - Hubbard 100 kV	Rebuild line with higher-capacity conductor	\$2,299,000	65% (\$1,500,000)
China - Ponnequin 100 kV Line	Rebuild line with higher-capacity conductor	\$4,790,000	62% (\$2,944,000)
Muldoon Mill - Hillcrest 100 kV line	Rebuild line with higher-capacity conductor	\$6,634,000	10% (\$688,000)
Bedford Rural - China 100 kV	Rebuild line with higher-capacity conductor	\$4,689,000	62% (\$2,882,000)
Wabash 500/230kV TX	Replace transformer with higher-rated unit	\$37,503,000	4% (\$1,660,000)
Adirondack Substation 500/230kV Auto	Replace transformer with higher-rated unit	\$36,411,000	2% (\$705,000)
Waynedale - Montpelier Electric Generating Station 230 kV	Rebuild line with higher-capacity conductor	\$71,072,000	2% (\$1,446,000)
Troy Energy Llc - Trego 500 kV	Rebuild line with higher-capacity conductor	\$186,104,000	1% (\$2,663,000)
James De Young - Adirondack 230 kV Line	Rebuild line with higher-capacity conductor	\$39,896,000	2% (\$692,000)
Pickford - Herald-Twin Cities 230 kV L	Rebuild line with higher-capacity conductor	\$45,163,000	2% (\$742,000)
E Methuen - Gray'S Flat 230 kV	Rebuild line with higher-capacity conductor	\$33,591,000	2% (\$564,000)
Mason St - New Departure 230 kV	Rebuild line with higher-capacity conductor	\$46,474,000	2% (\$760,000)
Adams - Onancock Diesel Generation Facility 230 kV	Rebuild line with higher-capacity conductor	\$67,517,000	2% (\$1,104,000)
Pickford - Adams 230 kV	Rebuild line with higher-capacity conductor	\$22,661,000	2% (\$376,000)
Trego - Beaver 500 kV	Reconductor line	\$197,466,000	2% (\$3,317,000)
Beaver 230-1 kV TX	Replace transformer with higher-rated unit	\$10,788,000	2% (\$188,000)
Beaver 500/1kV Transformer	Replace transformer with higher-rated unit	\$41,191,000	2% (\$719,000)
Gavin Power Llc 500-345 kV TX	Replace transformer with higher-rated unit	\$28,972,000	2% (\$515,000)
George Neal South - Myrle Point 230 kV	Reconductor line	\$4,724,000	2% (\$87,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Galeville - Gruver 230 kV	Reconductor line	\$3,792,000	2% (\$64,000)
Canal Road - Weedpatch 230 kV L	Reconductor line	\$3,320,000	2% (\$58,000)
Hubbard - Wrangell 100 kV	Reconductor line	\$810,000	65% (\$528,000)
	Total allocated to project		\$63,627,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.